Pricing Single-Name Credit Default Swaps

Seminar Module

conducted by

ACF Consultants Ltd





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Topics Covered

- Introduction to CDS pricing
- Credit risk and credit spreads
  - Term structure of credit spreads
  - Obtaining default probabilities from credit spreads
- Pricing Credit Default Swaps
  - Default and survival probabilities
  - Using default probabilities to obtain CDS premiums
  - Calculating up-front payment for CDS with standard premiums
  - The ISDA CDS Standard Model
  - Bootstrapping default probabilities from CDS premiums
  - Converting Between Par Spreads and Standard Premiums
  - Marking CDS to market
  - SDV01 and PV01
  - The Bloomberg CDSW screen

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Introduction to Credit Derivative Pricing

With so many factors driving the basis, asset swap prices are no longer a reliable way to price a CDS.

Let's take a different, fundamental approach based on **default probabilities**.

To price a credit derivative, we need the answers to two basic questions:

- What is the probability of a credit event?
- If a credit event occurs, how much is the protection seller likely to pay?

We may also need to know:

- If a credit event occurs, when does this happen?

Let's start by deriving a simple formula pricing CDS…

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CDS Pricing – A Simple Formula

Suppose that for a given entity we have determined:

- PD = 5%
- LGD = 60% (which implies a recovery rate RR of 40%)

Suppose we sell protection on 100 (N) identical such names, each on $1m (P)

- Expected number of losses = 5% × 100 = 5 (PD × N)
- Expected cost of losses = 5 × $1m × 60% = $3m (PD × N × P × LGD)

To offset these, we must collect a total of $3m of premium in total

- Premium per name = $30,000 (PD × N × P × LGD) / N
- Premium per name ≡ 300bp (PD × N × P × LGD) / (N × P)

CDS Premium = PD × LGD = PD × (1 – RR)

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Credit Risk and Credit Spreads

One way to obtain default probabilities is to observe credit spreads in the corporate bond market…

- Riskless investments establish a benchmark **riskless interest rate**
- Lenders and investors expect to receive a higher return from risky investments
- The difference between the risky and riskless rates is the

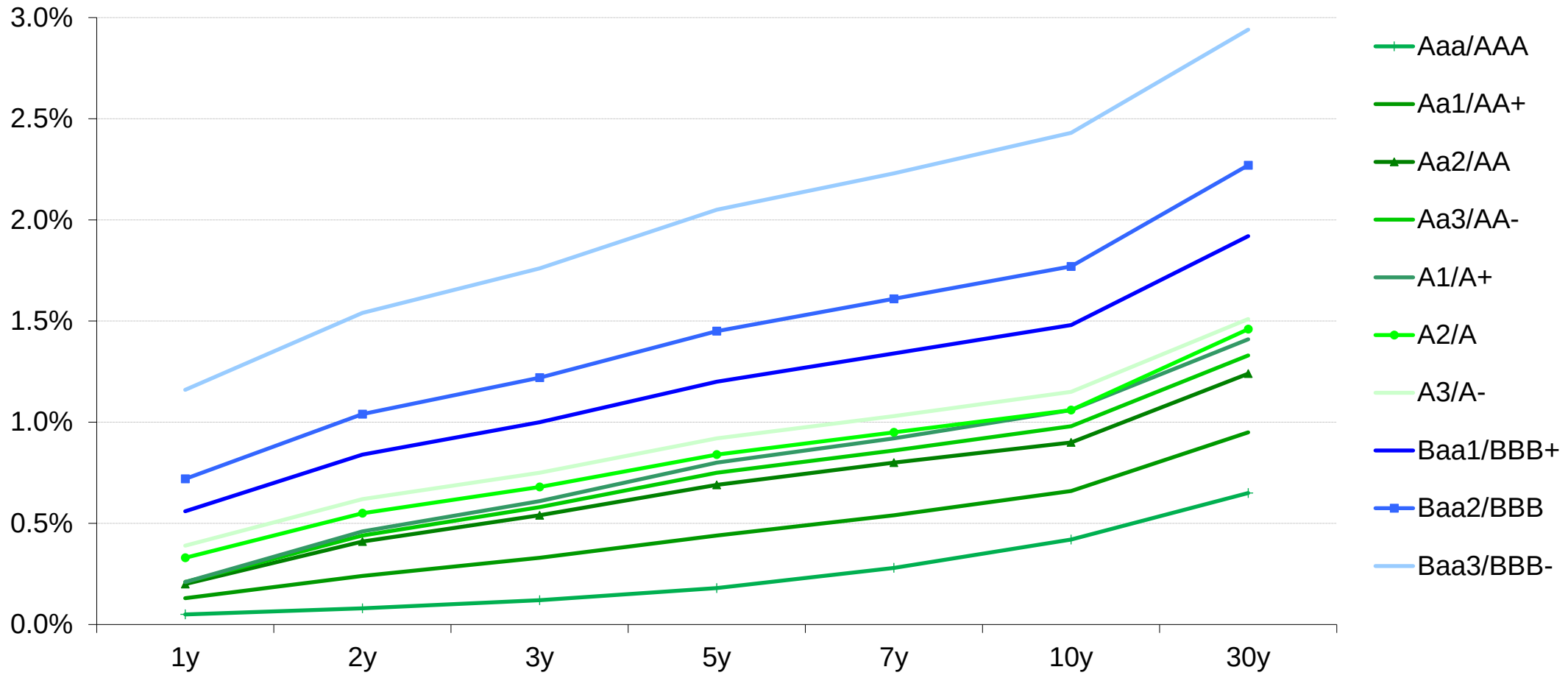
Credit Spread

- This will vary with:
  - Credit quality (e.g. credit rating)
  - Maturity

The charts overleaf provides an example of credit spreads…

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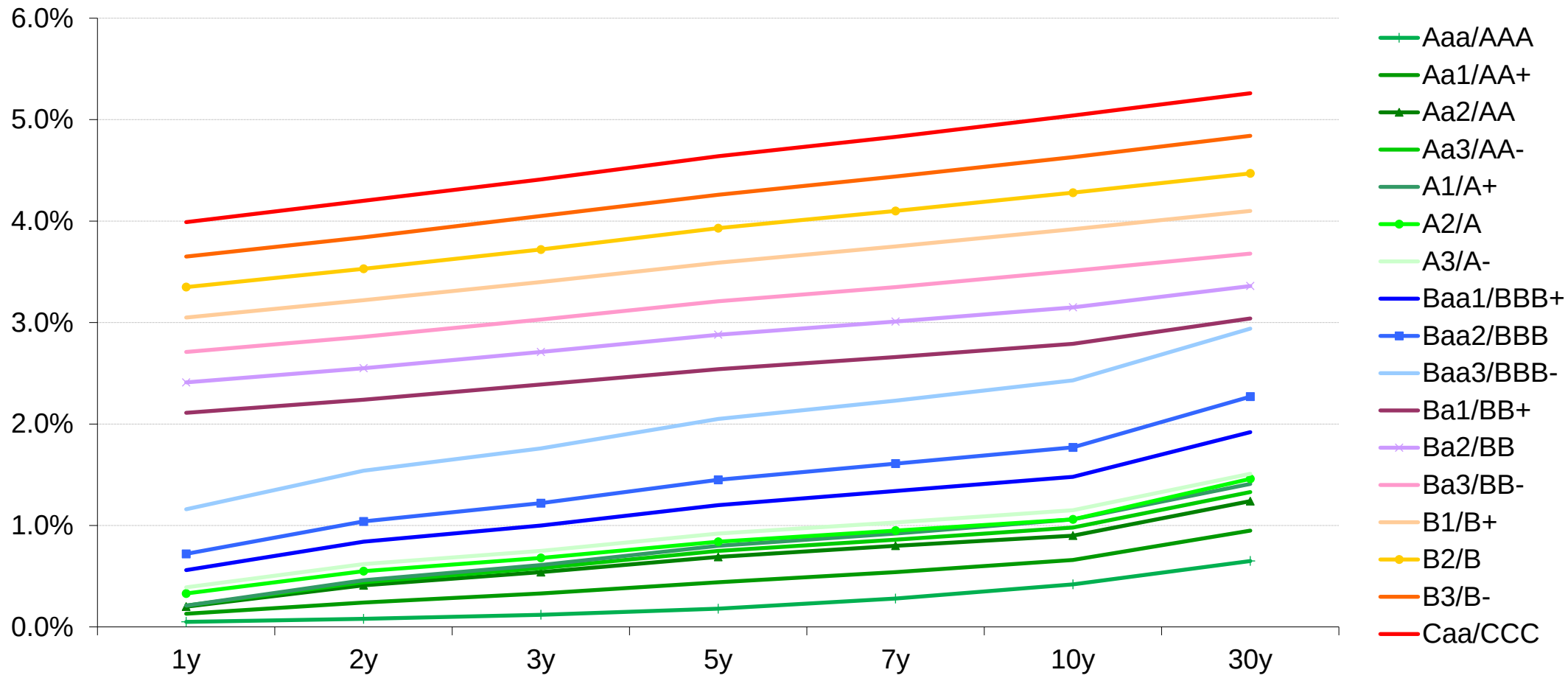
Example of Credit Spreads – Investment Grades



Source: Bonds Online 31st July 2015

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Example of Credit Spreads – All Grades



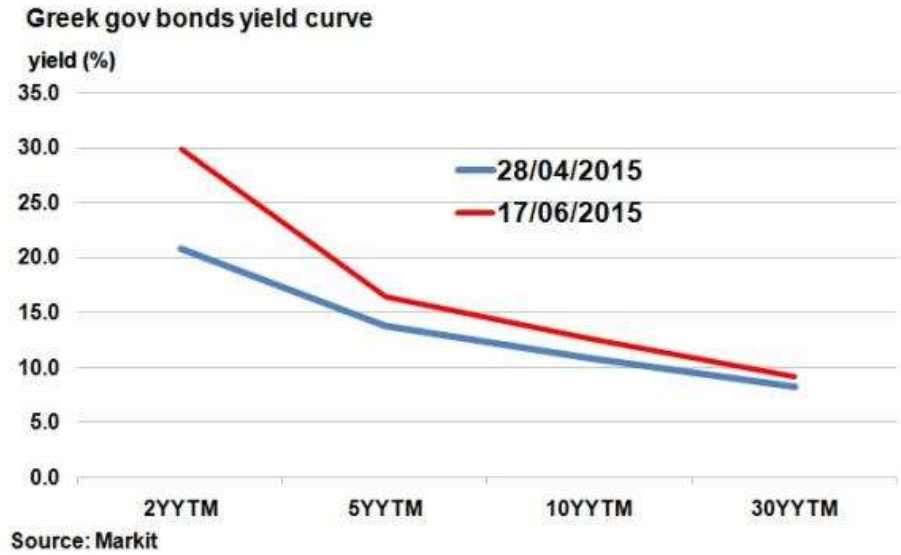
Source: Bonds Online 31st July 2015

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Term Structure of Credit Spreads

The examples clearly show that credit spreads depend on maturity. The **term structure of credit spreads** usually has these features:

- Lower-quality credits trade at a wider spread than higher-quality credits
- Longer-dated obligations normally have higher spreads than shorter-dated ones
  - 2yr AA: 40bp
  - 5yr AA: 54bp
  - 10yr AA: 80bp
- … but there can be exceptions, as shown opposite:
  - 2yr: 30%
  - 5yr: 16%
  - 30yr: 10%
- In the case of the credit curve in this chart:
  - Very high default risk on in June 2015 gave rise to high spread for short maturities
  - If Greece avoids default over the next few years, risk of later default is much lower, giving rise to lower spreads for longer maturities



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Default Probability and Corporate Bond Spreads

- Suppose that the corporate bonds of a particular issuer trade at the following yields*:

Maturity *t*	Risk-free rate *z*	Corporate bond yield *z+r*	Corporate spread *r*
1	4.00%	4.25%	0.25%
2	4.00%	4.50%	0.50%
3	4.00%	4.70%	0.70%
4	4.00%	4.85%	0.85%
5	4.00%	4.95%	0.95%

- The proceeds from investing in a risk-free asset are:

$$e^{zt}$$

- … while the proceeds from the risky asset are:

$$e^{(z+r)t}$$

- * Yields quoted at as continuously compounded rates

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Calculating Default Probabilities – Zero Recovery

- Let's assume that there is no recovery value following default
- If the probability of default is p, then an investor should be indifferent between an expected proceeds of:

$$(1-p)e^{(z+r)t} + p \times 0$$

- on the risky corporate bond (assuming no recovery), and:

$$e^{zt}$$

- …on the risk free bond. Setting these two expressions equal, and solving for *p* gives:

$$(1-p)e^{(z+r)t} = e^{zt}$$

$$\therefore (1-p)e^{rt} = 1$$

$$\therefore p = 1-e^{-rt}$$

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Calculating Default Probabilities – Examples

Using this formula…

$$p = 1 - e^{-rt}$$

We can therefore calculate the following probabilities of default from credit spreads:

Maturity *t*	Risk-free rate *z*	Corporate bond yield *z+r*	Corporate spread *r*	Cumulative probability of default	Annual probability of default
1	4.00%	4.25%	0.25%	0.24969%	0.24969%
2	4.00%	4.50%	0.50%	0.99502%	0.74533%
3	4.00%	4.70%	0.70%	2.07810%	1.08309%
4	4.00%	4.85%	0.85%	3.34285%	1.26475%
5	4.00%	4.95%	0.95%	4.63895%	1.29610%

e.g. $p_{0,5} = 1 - e^{-0.0095 \cdot 5} = 4.63895\%$ (cumulative probability of default over five-year period)

$p_{4,5} = p_{0,5} - p_{0,4} = 1.29610\%$ (probability of default over in year five)

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Calculating Default Probabilities with Recovery

- Now let's assume that the recovery rate following default is *R*.
- If the probability of default is *p*, then an investor should now be indifferent between an expected proceeds of:

$$(1-p)e^{(z+r)t} + pRe^{(z+r)t}$$

- On the risky corporate bond, and:

$$e^{zt}$$

- … on the risk-free bond. Again, setting these two expressions equal, and solving for *p* gives:

$$(1-p)e^{(z+r)t} + pRe^{(z+r)t} = e^{zt}$$

$$\therefore (1-p) + pR = e^{-rt}$$

$$\therefore p = \frac{1-e^{-rt}}{1-R}$$

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Calculating Default Probabilities – With Recovery

- Using this formula…

$$p = \frac{1 - e^{-rt}}{1 - R}$$

- … and assuming a recovery rate of 40%, we can calculate the following default probabilities:

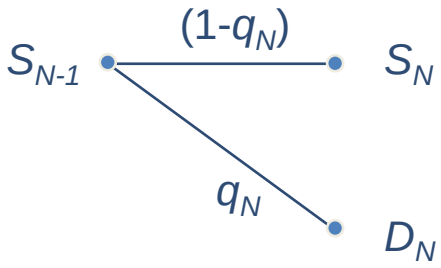
Maturity *t*	Risk-free rate *z*	Corporate bond yield *z+r*	Corporate spread *r*	Cumulative probability of default	Annual probability of default
1	4.00%	4.25%	0.25%	0.41615%	0.41615%
2	4.00%	4.50%	0.50%	1.65836%	1.24221%
3	4.00%	4.70%	0.70%	3.46351%	1.80514%
4	4.00%	4.85%	0.85%	5.57142%	2.10791%
5	4.00%	4.95%	0.95%	7.73159%	2.16017%

e.g. $p_{0,5} = \frac{1 - e^{-0.0095 \cdot 5}}{1 - 0.4} = 7.73159\%$ (cumulative probability of default over five-year period)

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Default and Survival Probabilities

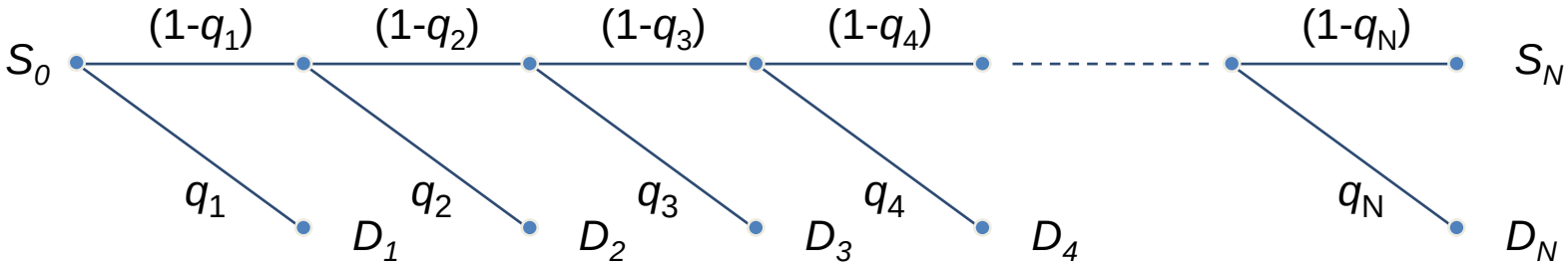
- Consider what happens to a risky asset over a specific period of time
- At each stage, having survived until that point, there are just two possibilities:
  - There is then a credit event, and the asset defaults
  - There is then no credit event, and the asset survives
- Let's call these outcomes:
  - D (for default) having probability q
  - S (for survival) having probability $(1-q)$
  - NB these probabilities are **conditional** upon having survived until that point
- We can represent this as a binary process:



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Multiple Periods

- Consider this binary process over multiple periods:



- The cumulative probability of survival to period N is then:

$$PS_N = (1-q_1) \times (1-q_2) \times (1-q_3) \times (1-q_4) \times \ldots \times (1-q_N) = PS_{N-1} \times (1-q_N)$$

- While the probability of default in any one period N is:

$$PD_N = PS_{N-1} \times q_N = PS_{N-1} - PS_N$$

- We can also link *PS* and *PD* by saying:

$$PS_N + \sum_{i=1}^{N} PD_i = 1$$

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Conditional Default Probabilities

On the previous page…

- PS_N and PD_N are probabilities **assessed at the outset**
- That's why the sum of all the PD_i plus PS_N must add to 1

$$PS_N + \sum_{i=1}^{N} PD_i = 1$$

- The q_i are **conditional probabilities**…
- For example, q_N is the probability of default in period *N* given that the entity has survived until period *N-1*
- The *q, PD*, and *PS* can be linked with the usual formula for conditional probability:

$$\mathrm{P}\left(D_N \middle| S_{N-1}\right) = q_N = \frac{PD_N}{PS_{N-1}}$$

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Default / Survival Probabilities – Simple Example

- Suppose the conditional default probability q_i each of the next two periods is 5%
- We can then calculate the following probabilities:

Period	PD	PS	q	1-q
0		100.00%		
1	5.00%	95.00%	5.00%	95.00%
2	4.75%	90.25%	5.00%	95.00%

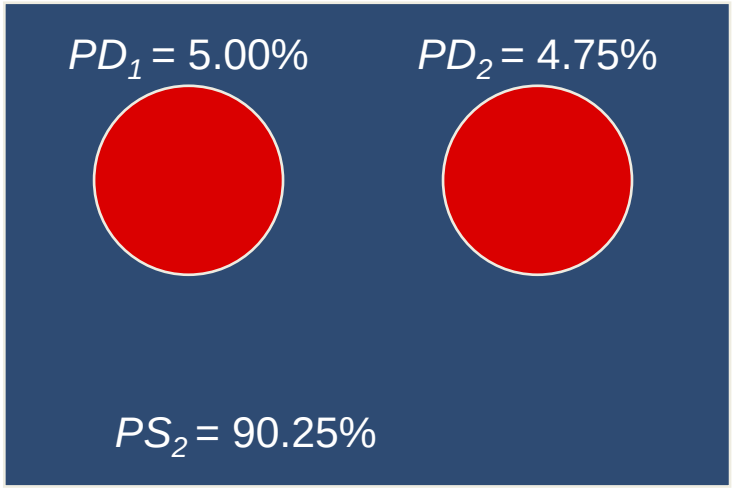
- The q's are conditional default probabilities
  - The probability of default in period 2 **measured at the outset** is: $PD_2 = 4.75\%$
  - The probability of default in period 2 **given survival through period 1** is: $q_2 = 5.00\%$
  - They are related by the formula for conditional probabilities: $q_2 = PD_2 / PS_1 = 4.75\% / 95\% = 5.00\%$
- The (1-q)'s are conditional survival probabilities
  - The probability of survival through period 2 **measured at the outset** is: $PS_2 = 90.25\%$
  - The probability of survival through period 2 **given survival through period 1** is: $1\text{-}q_2 = 95.00\%$
- The cumulative probability of survival after two periods (PS_2) is:

$$PS_2 = (1\text{-}q_1) \times (1\text{-}q_2) = 95.00\% \times 95.00\% = 90.25\%$$

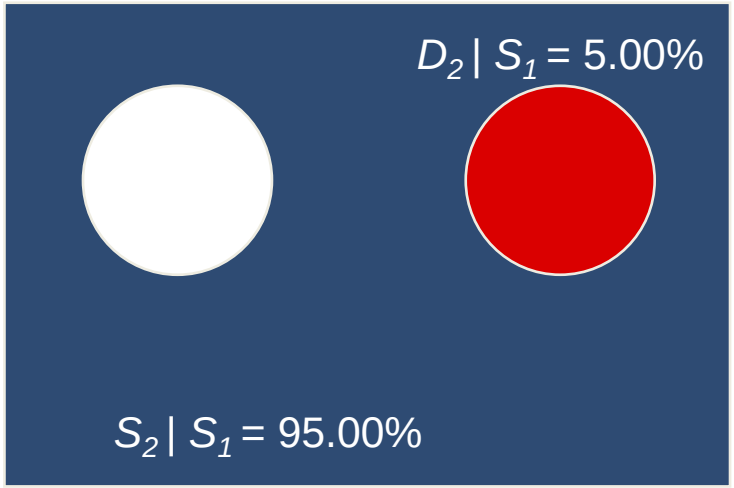
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Default / Survival Probabilities – Venn Diagrams

At the outset (t=0)



Following survival after period 1 (t=1)



Having explained the difference between the different kinds of probability, we will now focus on the *PD's* and *PS's.* This is because these are all measured at the outset, and we want to price a CDS at the outset.

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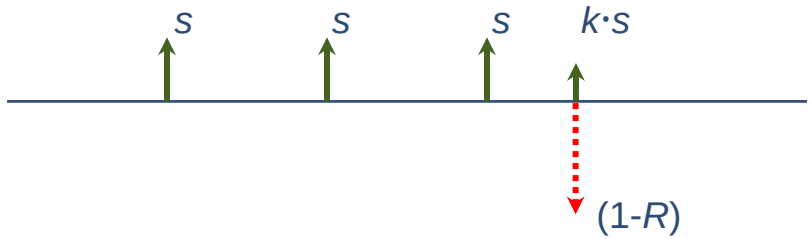
Pricing a Credit Default Swap

- Given a set of default probabilities, we can calculate the fair premium for a CDS
- To do this, consider a CDS as a series of contingent cash flows…
- … the cash flows depending upon whether a credit event occurs:

No Credit Event



Credit Event



where:

- *s* is the CDS premium
- *k* is the day count fraction when default occurred
- *R* is the recovery rate

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Valuing the Premium Income Stream



- The expected PV of the stream of CDS premiums over time can be calculated as:

$$PVS = s_N \sum_{j=1}^{N} T_{j-1,j} v_j PS_j$$

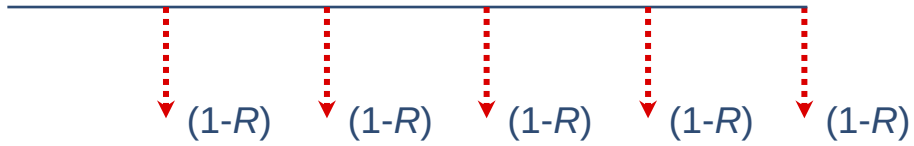
where:

- PVS is the expected present value of the stream of CDS premiums
- s_N is the CDS premium rate per annum for an N-period CDS
- v_j is the discount factor for period j
- PS_j is the probability of survival through period j
- $T_{j-1,j}$ is the length of time of period j (expressed as a fraction of a year)

- As we are using PS_j above, we are taking into account the fact that there may be a default at any time, so some of the premiums may not be received

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Valuing the Cost of Defaults



- If a default occurs in period *j*, the expected PV of the payout will be:

$$(1-R)v_j PD_j$$

- However, default can occur in **any** period, so we must sum this over time:

$$PVD = (1-R)\sum_{j=1}^{N} v_j PD_j$$

where:

- PVD is the expected PV of the cost of defaults over time
- v_j is the discount factor for period j
- PD_j is the probability of default in period j
- R is the recovery rate

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Determining the CDS Premium

- If a CDS is fairly priced, then the expected value of the premium stream must equal the expected value of the default payment. In other words, *PVS* = *PVD*
- We can therefore write:

$$s_N \sum_{j=1}^{N} T_{j-1,j} v_j PS_j = (1-R) \sum_{j=1}^{N} v_j PD_j$$

Expected PV of stream of premium receipts

Expected PV of cost of defaults over time

- Rearranging this expression gives an expression for the fair premium *s* for the CDS:

$$s_N = \frac{(1-R) \sum_{j=1}^{N} v_j PD_j}{\sum_{j=1}^{N} T_{j-1,j} v_j PS_j}$$

A

B

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Determining the CDS Premium – Example

To illustrate the pricing of a CDS, let's use the table of forward rates and survival probabilities in columns 2 and 4 to derive the set of CDS premiums shown in the last column (assuming a recovery rate of 40%, and a notional of $1):

Period	Forward rates	Discount factors v_j	Survival probability PS_j	Default probability PD_j	Probability-weighted PVs: Expected PV of receipt of 100% p.a. (B)	Probability-weighted PVs: Expected PV of default payment (A)	CDS Prem (bp) s_j
0.5	2.33%	0.9885	0.9305	0.0695	0.4599	0.0412	896
1.0	2.52%	0.9762	0.8656	0.0649	0.8824	0.0792	898
1.5	2.87%	0.9624	0.8151	0.0505	1.2746	0.1084	850
2.0	3.22%	0.9471	0.7730	0.0421	1.6407	0.1323	806
2.5	3.52%	0.9307	0.7432	0.0298	1.9865	0.1490	750
3.0	3.82%	0.9133	0.7145	0.0287	2.3128	0.1647	712
3.5	4.02%	0.8953	0.6990	0.0155	2.6257	0.1730	659
4.0	4.22%	0.8768	0.6837	0.0153	2.9254	0.1811	619
4.5	4.37%	0.8581	0.6806	0.0031	3.2174	0.1827	568
5.0	4.52%	0.8391	0.6776	0.0030	3.5017	0.1842	526

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CDS Premium – 1yr CDS

- On the previous page, the 1yr CDS premium is 898bp
- To see how this works out, we can calculate:
- Expected PV of premium on 6mth and 12mth dates:
  - Survival probability × Discount factor × Premium × Day count fraction
  - 6mth: 0.0898 × 0.5 × 0.9885 × 0.9305 = 0.0413
  - 12mth: 0.0898 × 0.5 × 0.9762 × 0.8656 = 0.0379
- Total expected present value of premium income = **0.0792**
- Expected PV of default payment if payment is made at end of period
  - Default probability × Discount factor × (1 – Recovery rate)
  - 6mth: (100% – 40%) × 0.9885 × 0.0695 = 0.0412
  - 12mth: (100% – 40%) × 0.9762 × 0.0649 = 0.0380
- Total expected present value of default payment = **0.0792**
- As these are the same, the 1yr CDS premium of 898bp must be correct!

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Calculating Up-Front Payments for Standard CDS

- From earlier, we have a formula for the expected PV of the stream of CDS premiums over time:

$$PVS = s_N \sum_{j=1}^{N} T_{j-1,j} v_j PS_j$$

- Suppose we collect the premium as a combination of:
  - An up-front payment PUF_N
  - A standard premium of PR_N
- The expected PV of this stream is simply:

$$PVS = PUF_N + PR_N \sum_{j=1}^{N} T_{j-1,j} v_j PS_j = s_N \sum_{j=1}^{N} T_{j-1,j} v_j PS_j$$

- So we can easily calculate the up-front payment as:

$$PUF_N = (s_N - PR_N) \sum_{j=1}^{N} T_{j-1,j} v_j PS_j = (1-R) \sum_{j=1}^{N} v_j PD_j - PR_N \sum_{j=1}^{N} T_{j-1,j} v_j PS_j$$

A C

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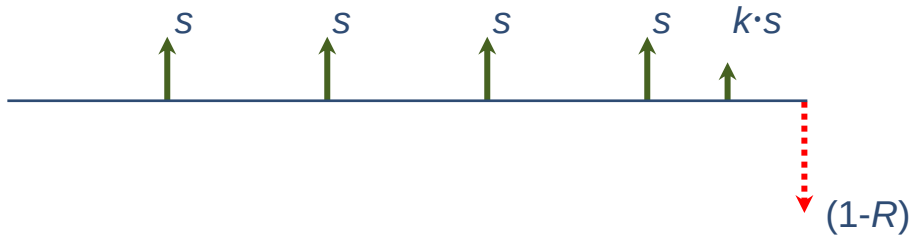
Calculating Up-Front Payments – Example

To illustrate the calculation of up-front payments, let's use the table of forward rates and survival probabilities in columns 2 and 4 this time to derive the set of up-front payments for a CDS with a standard premium of 500bp p.a.:

Period	Forward rates	Discount factors v_j	Survival probability PS_j	Default probability PD_j	Probability-weighted PVs: Expected PV of receipt of 500bp p.a. (C)	Probability-weighted PVs: Expected PV of default payment (A)	Up-front payment PUF_j
0.5	2.33%	0.9885	0.9305	0.0695	0.0230	0.0412	1.82%
1.0	2.52%	0.9762	0.8656	0.0649	0.0441	0.0792	3.51%
1.5	2.87%	0.9624	0.8151	0.0505	0.0637	0.1084	4.47%
2.0	3.22%	0.9471	0.7730	0.0421	0.0820	0.1323	5.03%
2.5	3.52%	0.9307	0.7432	0.0298	0.0993	0.1490	4.96%
3.0	3.82%	0.9133	0.7145	0.0287	0.1156	0.1647	4.90%
3.5	4.02%	0.8953	0.6990	0.0155	0.1313	0.1730	4.17%
4.0	4.22%	0.8768	0.6837	0.0153	0.1463	0.1811	3.48%
4.5	4.37%	0.8581	0.6806	0.0031	0.1609	0.1827	2.18%
5.0	4.52%	0.8391	0.6776	0.0030	0.1751	0.1842	0.91%

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Valuing the Premium Stream – Accrual Premium



If a default occurs half-way through period C, and the protection buyer pays a partial premium of $k \times s$, the expected PV of the fees received are now:

$$PVS_D = s_N \sum_{j=1}^{C-1} T_{j-1,j} v_j PS_j + s_N \frac{T_{C-1,C}}{2} v_C PD_C$$

and the value of the default payment is still:

$$(1-R) v_C PD_C$$

where:

- PVS_D is the expected PV of the stream of CDS premiums if there is default in period C
- PD_j is the probability of default in period j
- R is the recovery rate

… and other terms are as before

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Determining the CDS Premium – Accrual Premium

- If a CDS is fairly priced, then the expected value of the premium stream must equal the expected value of the default payment.
- As default can occur in any period *j*, we can therefore write:

$$s_N \sum_{j=1}^{N} T_{j-1,j} v_j PS_j + s_N \sum_{j=1}^{N} \frac{T_{j-1,j}}{2} v_j PD_j = (1-R) \sum_{j=1}^{N} v_j PD_j$$

Expected PV of stream of premium receipts	Expected PV of accrued premium payment in period when default occurs	Expected PV of cost of defaults over time

- Rearranging this expression gives the fair premium *s* for the CDS:

$$s_N = \frac{(1-R) \sum_{j=1}^{N} v_j PD_j}{\sum_{j=1}^{N} T_{j-1,j} v_j PS_j + \sum_{j=1}^{N} \frac{T_{j-1,j}}{2} v_j PD_j}$$

A, B, D

- This is the **ISDA CDS Standard Model**

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Determining the CDS Premium – Accrual Premium

The table below illustrates the pricing of a CDS including the accrual premium. The premiums are slightly lower than before because of the additional receipt of accrual premium:

B D A

Period	Forward rate	Discount factors v_j	Survival probability PS_j	Default probability PD_j	------ Probability-weighted PVs ------			CDS Premium (bp) s_j
					Expected PV of receipt of 100% p.a.	Expected PV of accrual premium	Expected PV of default payment	
0.5	2.33%	0.9885	0.9305	0.0695	0.4599	0.0172	0.0412	864
1.0	2.52%	0.9762	0.8656	0.0649	0.8824	0.0330	0.0792	866
1.5	2.87%	0.9624	0.8151	0.0505	1.2746	0.0452	0.1084	821
2.0	3.22%	0.9471	0.7730	0.0421	1.6407	0.0551	0.1323	780
2.5	3.52%	0.9307	0.7432	0.0298	1.9865	0.0621	0.1490	727
3.0	3.82%	0.9133	0.7145	0.0287	2.3128	0.0686	0.1647	692
3.5	4.02%	0.8953	0.6990	0.0155	2.6257	0.0721	0.1730	641
4.0	4.22%	0.8768	0.6837	0.0153	2.9254	0.0754	0.1811	603
4.5	4.37%	0.8581	0.6806	0.0031	3.2174	0.0761	0.1827	555
5.0	4.52%	0.8391	0.6776	0.0030	3.5017	0.0767	0.1842	515

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CDS Premium – 1yr CDS with Accrual Premium

- On the previous page, the 1yr CDS premium is 866bp
- To see how this works out, we can calculate:
- Expected PV of premium on 6mth and 12mth dates:
  - Survival probability × Discount factor × Premium × Day count fraction
  - 6mth: 0.0866 × 0.5 × 0.9885 × 0.9305 = 0.0398
  - 12mth: 0.0866 × 0.5 × 0.9762 × 0.8656 = 0.0366
- Expected PV of accrued premium if default occurs half-way through a period
  - Default probability × Discount factor × Premium × Day count fraction
  - 6mth: 0.0866 × 0.25 × 0.9885 × 0.0695 = 0.0015
  - 12mth: 0.0866 × 0.25 × 0.9762 × 0.0649 = 0.0014
- Total expected present value of premium income = **0.0792**
- Expected PV of default payment if payment is made at end of period
  - Default probability × Discount factor × (1 – Recovery rate)
  - 6mth: (100% – 40%) × 0.9885 × 0.0695 = 0.0412
  - 12mth: (100% – 40%) × 0.9762 × 0.0649 = 0.0380
- Total expected present value of default payment = **0.0792**

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Linking Default Probabilities and CDS Premiums

- The previous formulas solve for the CDS premium given the default probabilities
- But which comes first?
  - Default probabilities, or
  - CDS premiums?
- We have so far started with default probabilities, and then determined CDS premiums
- This assumes that we can obtain reliable default probabilities…
  - … from bond spreads
  - … or from cumulative default probability tables published by the credit rating agencies
- In practice, it is the CDS market that is the most liquid!
- So, in terms of this “chicken and egg” problem:
  - CDS premiums come first
  - From these you can infer default probabilities

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Default Probabilities from CDS Premiums

The formula given previously solves for the *N*-period CDS premium s_N in terms of:

- Default probabilities *PD*
- Survival probabilities *PS*
- Discount factors *v*
- The day-count fraction *T*

$$s_N = \frac{(1-R)\sum_{j=1}^{N} v_j PD_j}{\sum_{j=1}^{N} T_{j-1,j} v_j PS_j}$$

- This formula can be re-arranged to solve for the default probability *PD* given the CDS premium s:

$$PD_N = \frac{s_N\left[\sum_{j=1}^{N-1} v_j T_{j-1,j} PS_j + v_N T_{N-1,N} PS_{N-1}\right] - (1-R)\sum_{j=1}^{N-1} v_j PD_j}{v_N\left[s_N T_{N-1,N} + (1-R)\right]}$$

- Let's examine this formula more closely…

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Analysing the Formula for Default Probability

- The formula for determining *s* from the default and survival probabilities *PD* and *PS* was quite straightforward…
  - *PD* and *PS* (the data) is on the right
  - *s* (what we are trying to solve for) is on the left
- The new formula appears to be a little mixed-up!
  - *s* (the data) is on the right
  - *PD* and *PS* (what we are trying to solve for) is BOTH on the left and the right

$$PD_N = \frac{s_N\left[\sum_{j=1}^{N-1} v_j T_{j-1,j} PS_j + v_N T_{N-1,N} PS_{N-1}\right] - (1-R)\sum_{j=1}^{N-1} v_j PD_j}{v_N\left[s_N T_{N-1,N} + (1-R)\right]}$$

- In fact, this is an example of a **bootstrapping** formula
  - To solve for PD_N, you need s_N, as well as PD_{N-1} and PS_{N-1}
  - To solve for PD_{N-1}, you need s_{N-1}, as well as PD_{N-2} and PS_{N-2} … and so on
- Let's look at a worked example

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Bootstrapping Default Probabilities – Step 1

- First calculate PD_1:

  Suppose:

  - s_1 = 896bp
  - v_1 = 0.9885
  - R = 40%
  - $T_{0,1}$ = 0.5

$$PD_1 = \frac{s_1\left[v_1 T_{0,1} PS_0\right]}{v_1\left[s_1 T_{0,1} + (1-R)\right]} = \frac{0.0896\left[0.9885 \times 0.5 \times 1\right]}{0.9885\left[0.0896 \times 0.5 + (1-0.4)\right]} = 0.0695$$

- Of course, the one-period survival probability PS_1 is:

$$PS_1 = 1 - PD_1 = 1 - 0.0695 = 0.9305$$

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Bootstrapping Default Probabilities – Step 2

- Now we have PD_1, we can calculate PD_2:

  Suppose:

  - s_2 = 898bp
  - R = 40%
  - v_2 = 0.9762
  - $T_{1,2}$ = 0.5

  And from the previous calculation:

  - PD_1 = 0.0695
  - PS_1 = 0.9305

$$PD_2 = \frac{s_2\left[\sum_{j=1}^{1} v_j T_{j-1,j} PS_j + v_2 T_{1,2} PS_1\right] - (1-R)\sum_{j=1}^{1} v_j PD_j}{v_2\left[s_2 T_{1,2} + (1-R)\right]}$$

$$= \frac{0.0898\left[0.9885 \times 0.5 \times 0.9305 + 0.9762 \times 0.5 \times 0.9305\right] - (1-0.4)0.9885 \times 0.0695}{0.9762\left[0.0898 \times 0.5 + (1-0.4)\right]}$$

$$= 0.0649$$

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Bootstrapping Default Probabilities – Step 3

- We therefore continue the process
- As we calculate each default probability, we can use this to calculate the next default probability…
- … until the table of default and survival probabilities is complete

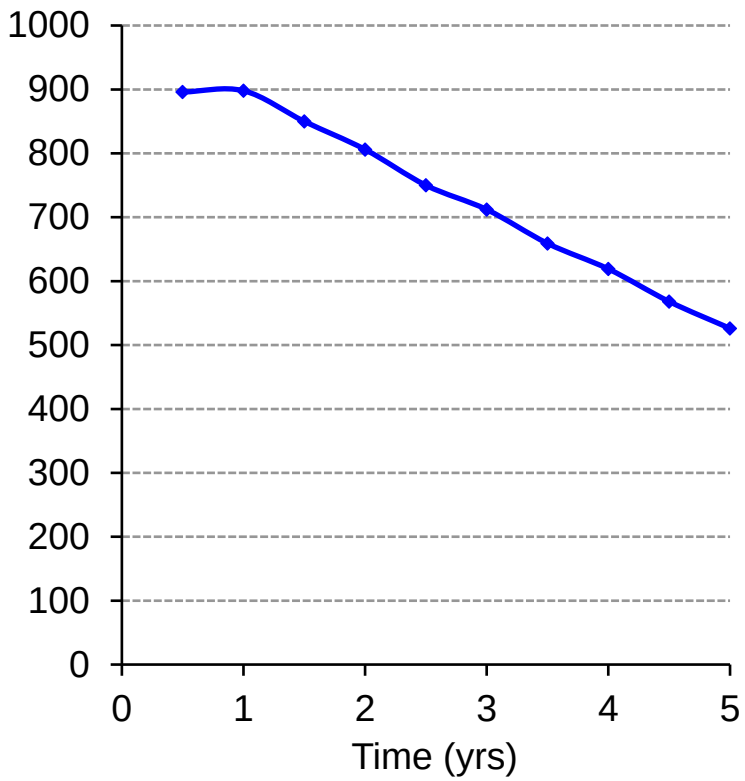
Period	Forward rates	Discount factors v_j	CDS Premium (bp) s_j	Default probability PD_j	Survival probability PS_j
0.5	2.33%	0.9885	896	0.0695	0.9305
1.0	2.52%	0.9762	898	0.0649	0.8656
1.5	2.87%	0.9624	850	0.0505	0.8151
2.0	3.22%	0.9471	806	0.0421	0.7730
2.5	3.52%	0.9307	750	0.0298	0.7432
3.0	3.82%	0.9133	712	0.0287	0.7145
3.5	4.02%	0.8953	659	0.0155	0.6990
4.0	4.22%	0.8768	619	0.0153	0.6837
4.5	4.37%	0.8581	568	0.0031	0.6806
5.0	4.52%	0.8391	526	0.0030	0.6776

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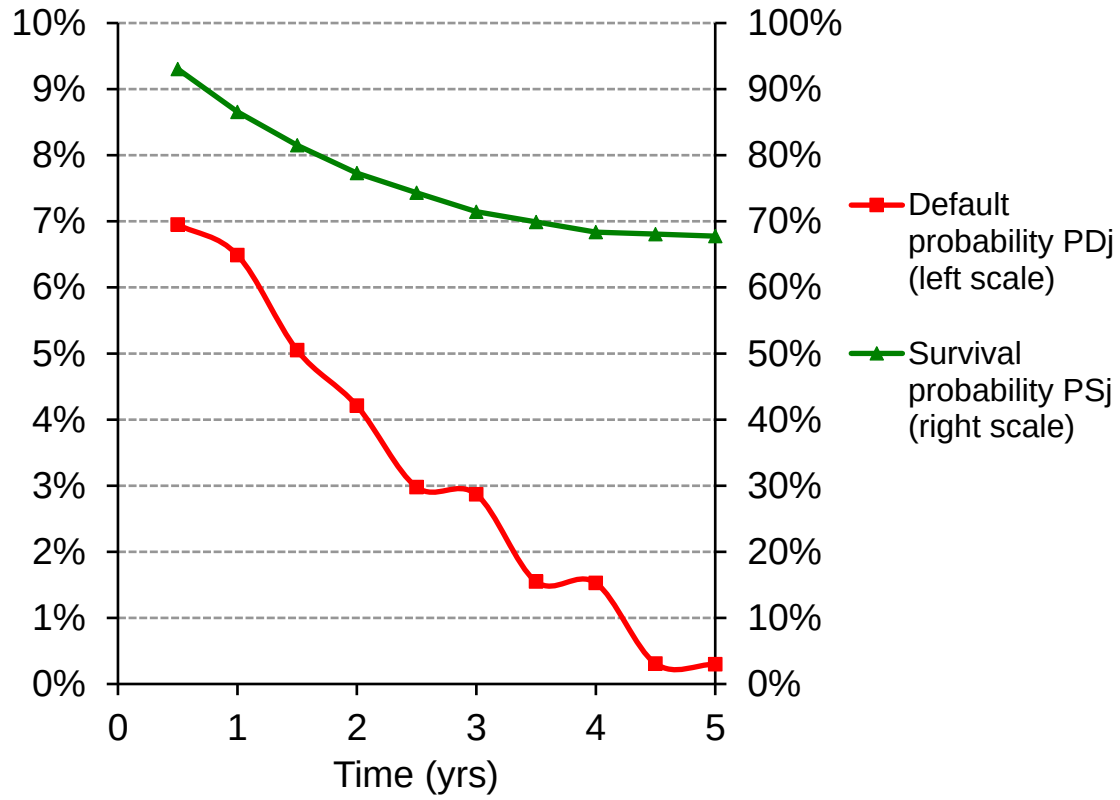
Term Structure of Default Probabilities

From the example on the previous page, we can construct a term structure of default probabilities…

CDS Premium (bp)



Default and Survival Probabilities



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Approximating Default and Survival Probabilities

There are some approximations to the formulas given earlier:

- One-Period Default Probability

$$PD \simeq \frac{s}{1-R} \quad e.g.\ PD_{6mth} = \frac{896bp/2}{1-0.4} = 7.5\%$$

  - This compares to the exact answer of 7.0%

- Term Survival Probability

$$PS_N \simeq \left(1-\frac{s_N}{1-R}\right)^N \quad e.g.\ PS_{5yr} = \left(1-\frac{526bp/2}{1-0.4}\right)^{10} = 63.9\%$$

  - This compares to the exact answer of 67.8%

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Converting Par Spreads to Standard Premiums

- Earlier, we calculated up-front payment using the probabilities PS_j and PD_j
- In practice, we can't observe these directly...
- What we do observe are CDS quotations in the form of par spreads
- To convert from par spreads to standard premiums, we therefore need to:
  - Bootstrap the set of default and survival probabilities from the par spreads, as explained before
  - Calculate the up-front payment using the formula we saw before:

$$PUF_N = (s_N - PR_N)\sum_{j=1}^{N} T_{j-1,j} v_j PS_j = (1-R)\sum_{j=1}^{N} v_j PD_j - PR_N \sum_{j=1}^{N} T_{j-1,j} v_j PS_j$$

A C

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Converting Par Spreads to Standard Premiums

In the example below, we first bootstrap the set of default and survival probabilities from the set of CDS par spreads in column 4, and then convert these into a standard premium plus an up-front payment (assuming standard premium of 500bp p.a., a recovery rate of 40%, and a notional of $1):

C

A

Period	Forward rates	Discount factors v_j	CDS Premium s_j	Survival probability PS_j	Default probability PD_j	Probability-weighted PVs		Up-front payment PUF_j
						Expected PV of receipt of 500bp p.a.	Expected PV of default payment	
0.5	2.33%	0.9885	3600	0.7692	0.2308	0.0190	0.1369	11.79%
1.0	2.52%	0.9762	3400	0.6095	0.1597	0.0339	0.2304	19.65%
1.5	2.87%	0.9624	3100	0.5124	0.0972	0.0462	0.2865	24.03%
2.0	3.22%	0.9471	2800	0.4550	0.0574	0.0570	0.3191	26.21%
2.5	3.52%	0.9307	2500	0.4272	0.0278	0.0669	0.3346	26.77%
3.0	3.82%	0.9133	2200	0.4230	0.0043	0.0766	0.3370	26.04%
3.5	4.02%	0.8953	2100	0.3842	0.0387	0.0852	0.3578	27.26%
4.0	4.22%	0.8768	2000	0.3571	0.0271	0.0930	0.3720	27.90%
4.5	4.37%	0.8581	1900	0.3395	0.0176	0.1003	0.3811	28.08%
5.0	4.52%	0.8391	1800	0.3298	0.0096	0.1072	0.3860	27.87%

CDS premiums are those of Countrywide Financial on 8th Jan 2008, just before they were taken over by Bank of America. Note the relatively flat up-front curve beyond one year.

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Converting Standard Premiums to Par Spread

To convert from standard premium with up-front payment to par spread:

- Bootstrap the set of default and survival probabilities using:

$$PD_N = \frac{PR_N\left[\sum_{j=1}^{N-1} v_j T_{j-1,j} PS_j + v_N T_{N-1,N} PS_{N-1}\right] - (1-R)\sum_{j=1}^{N-1} v_j PD_j + PUF_N}{v_N\left[PR_N T_{N-1,N} + (1-R)\right]}$$

(this is a modified version of the bootstrap formula explained earlier)

- Then calculate the par spread using:

$$s_N = \frac{(1-R)\sum_{j=1}^{N} v_j PD_j}{\sum_{j=1}^{N} T_{j-1,j} v_j PS_j}$$

A

B

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Converting Standard Premiums to Par Spread

In the example below, we first bootstrap the set of default and survival probabilities from the set of CDS up-front payments in column 4 (assuming standard premium of 500bp p.a.), and then use these probabilities to calculate par spreads (assuming a recovery rate of 40%, and a notional of $1):

B

A

Period	Forward rates	Discount factors v_j	Up-front payment PUF_j	Survival probability PS_j	Default probability PD_j	Probability-weighted PVs		CDS Premium s_j
						Expected PV of receipt of 100% p.a.	Expected PV of default payment	
0.5	2.33%	0.9885	11.79%	0.7692	0.2308	0.3802	0.1369	3600
1.0	2.52%	0.9762	19.65%	0.6095	0.1597	0.6777	0.2304	3400
1.5	2.87%	0.9624	24.03%	0.5124	0.0972	0.9242	0.2865	3100
2.0	3.22%	0.9471	26.21%	0.4550	0.0574	1.1397	0.3191	2800
2.5	3.52%	0.9307	26.77%	0.4272	0.0278	1.3385	0.3346	2500
3.0	3.82%	0.9133	26.04%	0.4230	0.0043	1.5317	0.3370	2200
3.5	4.02%	0.8953	27.26%	0.3842	0.0387	1.7036	0.3578	2100
4.0	4.22%	0.8768	27.90%	0.3571	0.0271	1.8602	0.3720	2000
4.5	4.37%	0.8581	28.08%	0.3395	0.0176	2.0058	0.3811	1900
5.0	4.52%	0.8391	27.87%	0.3298	0.0096	2.1442	0.3860	1800

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Marking CDS to Market

- When the CDS curve changes, existing CDS contracts will exhibit a mark-to-market profit or loss
- To determine this:
  - Use the current CDS quotes to construct a new set of default and survival probabilities
  - Use the current set of interest rates to construct a new set of discount factors
  - Value the stream of swap payments and default payments with these new curves
  - The difference between the value of these two legs is the mark-to-market profit

The example on the following pages illustrates this…

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Mark to Market Example – Starting Point

- Suppose that investors buy CDSs at the range of premiums shown in column 3
- If market rates don't change, all the CDSs are fair-valued with an NPV of zero

	Discount factors	CDS Premium	Survival probability	Default probability	Expected PV of premium payment	Exp PV of contingent receipt	NPV for buyer on $1m
0.5	0.9885	896	0.9305	0.0695	41,220	41,220	0
1	0.9762	898	0.8656	0.0649	79,232	79,232	0
1.5	0.9624	850	0.8151	0.0505	108,392	108,392	0
2	0.9471	806	0.7730	0.0421	132,317	132,317	0
2.5	0.9307	750	0.7432	0.0298	148,958	148,958	0
3	0.9133	712	0.7145	0.0287	164,685	164,685	0
3.5	0.8953	659	0.6990	0.0155	173,012	173,012	0
4	0.8768	619	0.6837	0.0153	181,061	181,061	0
4.5	0.8581	568	0.6806	0.0031	182,657	182,657	0
5	0.8391	526	0.6776	0.0030	184,167	184,167	0

Let's now explore what happens if CDS premiums rise…

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Mark to Market Example – Rates Rise

- In the example below, the entire credit curve has risen by 1%
- The table recalculates the default and survival probabilities, and then the mark-to-market profits, on a notional of $1m

	Discount factors	Old CDS Premium	New CDS Premium	Survival probability	Default probability	Expected PV of premium payment	Exp PV of contingent receipt	NPV for buyer on $1m
0.5	0.9885	896	996	0.9233	0.0767	40,903	45,466	4,564
1	0.9762	898	998	0.8523	0.0710	78,333	87,057	8,724
1.5	0.9624	850	950	0.7964	0.0559	106,777	119,333	12,556
2	0.9471	806	906	0.7495	0.0469	129,888	145,993	16,105
2.5	0.9307	750	850	0.7152	0.0343	145,723	165,157	19,434
3	0.9133	712	812	0.6824	0.0328	160,570	183,120	22,550
3.5	0.8953	659	759	0.6628	0.0196	168,136	193,654	25,517
4	0.8768	619	719	0.6437	0.0192	175,395	203,734	28,339
4.5	0.8581	568	668	0.6364	0.0072	176,384	207,453	31,069
5	0.8391	526	626	0.6294	0.0070	177,292	211,002	33,710

The resultant profit on the 5yr swap is $33,710.

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Calculating the SDV01

- To calculate the SDV01 (sensitivity of a CDS to a 1bp shift in CDS premiums)
  - Shift the credit curve by 1bp
  - Recalculate the default and survival probabilities
  - Then calculate the mark-to-market profits, on a notional of $1m

	Discount factors	Old CDS Premium	New CDS Premium	Survival probability	Default probability	Expected PV of premium payment	Exp PV of contingent receipt	NPV for buyer on $1m
0.5	0.9885	896	897	0.9304	0.0696	41,217	41,263	46
1	0.9762	898	899	0.8655	0.0650	79,223	79,312	88
1.5	0.9624	850	851	0.8149	0.0506	108,376	108,503	127
2	0.9471	806	807	0.7728	0.0421	132,292	132,456	164
2.5	0.9307	750	751	0.7429	0.0298	148,926	149,124	199
3	0.9133	712	713	0.7142	0.0287	164,643	164,875	231
3.5	0.8953	659	660	0.6986	0.0155	172,962	173,224	262
4	0.8768	619	620	0.6833	0.0153	181,003	181,295	292
4.5	0.8581	568	569	0.6801	0.0031	182,592	182,914	322
5	0.8391	526	527	0.6771	0.0030	184,096	184,446	350

The P&L for a 100bp move in CDS premiums of $33,710 is a little less than 100 times the SDV01 of $350, because the SDV01 reduces as CDS premiums rise.

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Mark to Market with Standardized Premiums – Starting Point

- In a world of standardized premiums, there will be an up-front payment (as shown in column 7)
- As before, all the CDSs are fair-valued with an NPV of zero

	Discount factors	Standard Premium	Survival probability	Default probability	Expected PV of premium payment	Upfront Payment	Exp PV of contingent receipt	NPV for buyer on $1m
0.5	0.9885	500	0.9305	0.0695	22,995	18,225	41,220	0
1	0.9762	500	0.8656	0.0649	44,119	35,113	79,232	0
1.5	0.9624	500	0.8151	0.0505	63,730	44,662	108,392	0
2	0.9471	500	0.7730	0.0421	82,033	50,284	132,317	0
2.5	0.9307	500	0.7432	0.0298	99,326	49,632	148,958	0
3	0.9133	500	0.7145	0.0287	115,640	49,045	164,685	0
3.5	0.8953	500	0.6990	0.0155	131,286	41,726	173,012	0
4	0.8768	500	0.6837	0.0153	146,272	34,788	181,061	0
4.5	0.8581	500	0.6806	0.0031	160,872	21,785	182,657	0
5	0.8391	500	0.6776	0.0030	175,087	9,081	184,167	0

Let's explore again what happens if CDS premiums rise…

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Mark to Market with Standardized Premiums – Rates Rise

- As before, the entire credit curve has risen by 1%, and the default and survival probabilities change
- The mark-to-market profits are almost identical to those calculated previously

	Discount factors	Standard Premium	Survival probability	Default probability	Expected PV of premium payment	Upfront Payment	Exp PV of contingent receipt	NPV for buyer on $1m
0.5	0.9885	500	0.9233	0.0767	22,818	18,225	45,466	4,423
1	0.9762	500	0.8523	0.0710	43,618	35,113	87,057	8,325
1.5	0.9624	500	0.7964	0.0559	62,780	44,662	119,333	11,890
2	0.9471	500	0.7495	0.0469	80,527	50,284	145,993	15,182
2.5	0.9307	500	0.7152	0.0343	97,169	49,632	165,157	18,356
3	0.9133	500	0.6824	0.0328	112,751	49,045	183,120	21,325
3.5	0.8953	500	0.6628	0.0196	127,586	41,726	193,654	24,341
4	0.8768	500	0.6437	0.0192	141,695	34,788	203,734	27,250
4.5	0.8581	500	0.6364	0.0072	155,347	21,785	207,453	30,321
5	0.8391	500	0.6294	0.0070	168,550	9,081	211,002	33,371

The $33,371 profit on the 5yr swap is almost identical with the $33,710 profit calculated previously.

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Formula for SDV01

- The example on the previous page applied to a CDS with premiums at the current par spread s_N, and therefore starting with a zero NPV. In general:
  - CDS premiums are executed at a standardised premium *p*, and have an initial non-zero NPV
  - The SDV01 is then the difference between the NPV at s_N and the NPV at s'_N, 1bp higher:

$$SDV01 = \left[(1-R) \sum_{j=1}^{N} v_j PD'_j - p \sum_{j=1}^{N} T_{j-1,j} v_j PS'_j \right] - \left[(1-R) \sum_{j=1}^{N} v_j PD_j - p \sum_{j=1}^{N} T_{j-1,j} v_j PS_j \right]$$

where:
- *PDj* and *PSj* are the set of default and survival probabilities at market level s_N
- *PD'j* and *PS'j* are the set of default and survival probabilities at market level s'_N, 1bp higher
- The notional principal is *$1*

- From the standard formula for pricing a CDS (developed earlier):

$$(1-R) \sum_{j=1}^{N} v_j PD_j = s_N \sum_{j=1}^{N} T_{j-1,j} v_j PS_j \quad \text{and} \quad (1-R) \sum_{j=1}^{N} v_j PD'_j = (s_N + 1\text{bp}) \sum_{j=1}^{N} T_{j-1,j} v_j PS'_j$$

- On the next page, we substitute these expressions into the first formula…

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SDV01 vs. PV01

- From the previous page we have:

$$SDV01 = (s_N + 1\text{bp} - p)\sum_{j=1}^{N} T_{j-1,j} v_j PS'_j - (s_N - p)\sum_{j=1}^{N} T_{j-1,j} v_j PS_j$$

- Note that, in general, the SDV01 differs from the PV01 of a CDS

$$PV01 = (s_N - (p - 1\text{bp}))\sum_{j=1}^{N} T_{j-1,j} v_j PS_j - (s_N - p)\sum_{j=1}^{N} T_{j-1,j} v_j PS_j = 1\text{bp} \times \sum_{j=1}^{N} T_{j-1,j} v_j PS_j$$

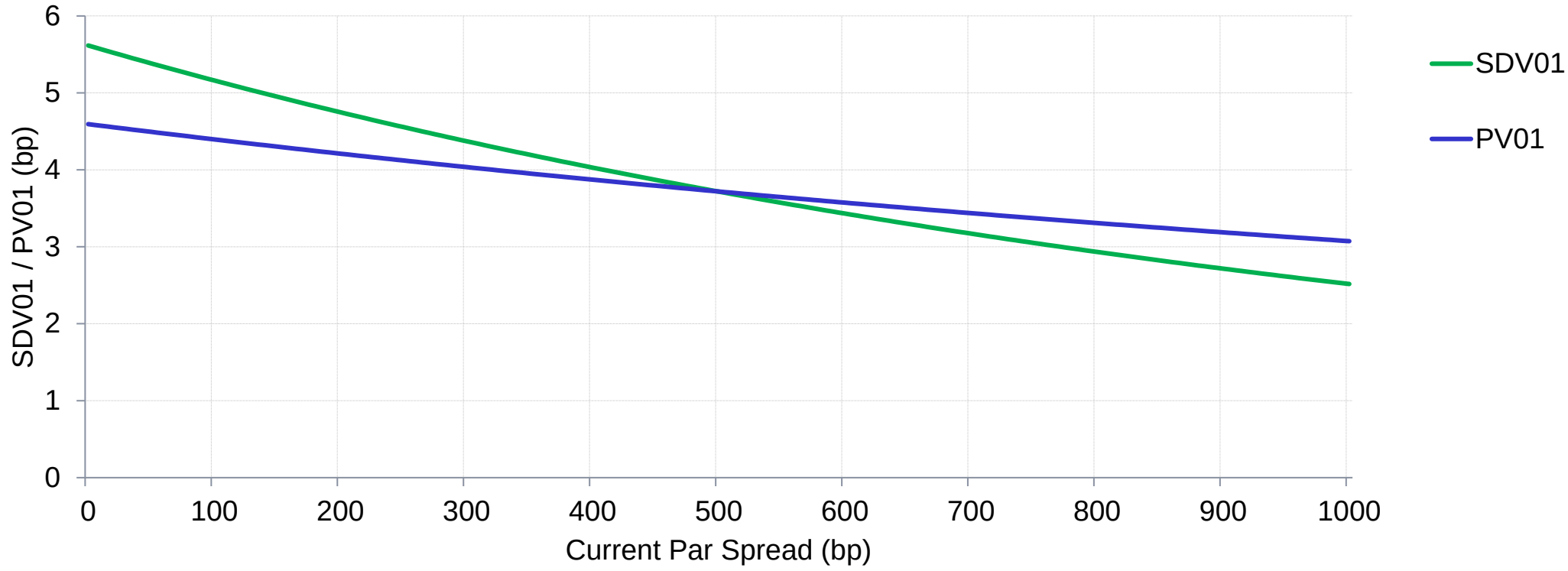
- ***SDV01*** is the change in NPV caused by a 1bp change in the *current* credit curve on a CDS executed at *historical* (off-market) rates.
- ***PV01*** is the change in NPV caused by a 1bp change in the CDS premium valued at *current* market rates. It is also known as the ***risky PV***.
- Example:
  - Original par spread: 300bp
  - Current par spread: 350bp
  - Standard premium: 500bp

SDV01 = 4.20bp
PV01 = 3.95bp

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SDV01 and PV01 Compared

- In general, the SDV01 will differ from the PV01, as the chart below shows…
- SDV01 and PV01 are only identical when s_N + 1bp = p



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The Bloomberg CDSW Screen



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